

Question 1: Sampling and Central Limit Theorem

In this question we're given three distributions: $\text{Poi}(10)$, $\text{Exp}(1/2)$, $\text{Geo}(1/5)$. We'll see about the CLT by experiment and will get some intuition about it.

Part 1: Expectations and Variances of The Distributions

For Poisson distribution with parameter λ the expectation is equal to λ (which is $n \cdot p$ actually) and the variance equals λ too (which is also $n \cdot p \cdot (1-p) = n \cdot p$).

So: If $X \sim \text{Poi}(10)$ Then: $E[X] = 10$ and $\text{Var}(X) = 10$

In exponential distribution, expectation equals $1/\lambda$ and variance is equal to $1/\lambda^2$. So:

If $Y \sim \text{Exp}(1/2)$: $E[Y] = 2$ and $\text{Var}(Y) = 1 / (1/4) = 4$

Also for geometric distribution the expectation is $1/p$ and the variance is $(1-p) / p^2$.

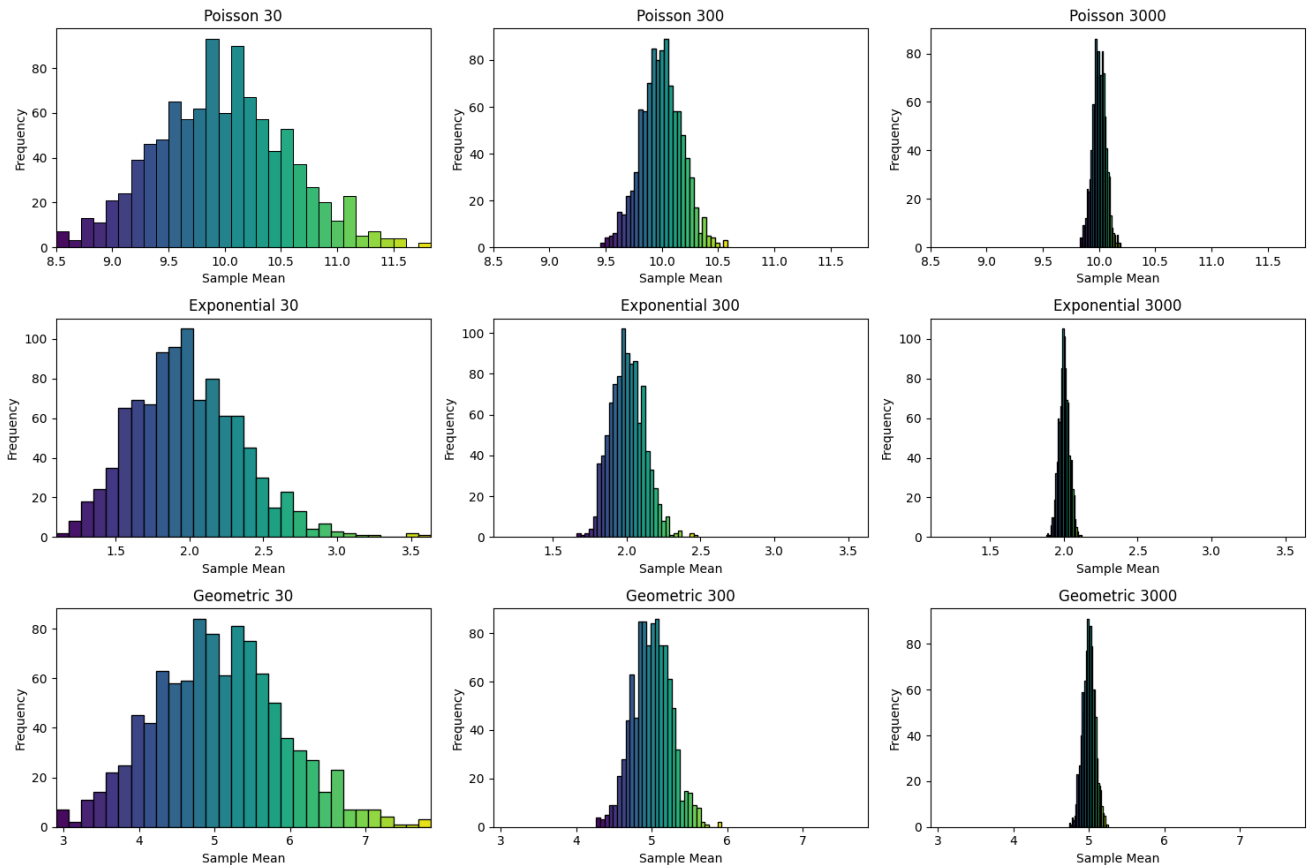
If $Z \sim \text{Geo}(1/5)$: $E[Z] = 5$ and $\text{Var}(Z) = (1 - 1/5) / (1/25) = 20$

Part 2: Sampling and Plotting Means

First I have my samples ready and calculate their means in-place by Numpy built-in functions.

Note: For exponential distribution sampling `np.random.exponential()` function takes the scale parameter which is equal to the expectation of the distribution that is $1 / (1/2) = 2$.

Then I plot the data by histograms in matplotlib.



Part 3: Analysing Mean and Std of Samplings

I'll simply compose some Numpy functions and print the means and stds of the samplings.

```
***** Poisson *****
Actual mean: 10
Actual std: 3.16228
The mean of mean of 1000 Poisson samplings when sample size = 30 is equal to: 10.01557
The std of mean of 1000 Poisson samplings when sample size = 30 is equal to: 0.57042
The mean of mean of 1000 Poisson samplings when sample size = 300 is equal to: 9.99367
The std of mean of 1000 Poisson samplings when sample size = 300 is equal to: 0.18540
The mean of mean of 1000 Poisson samplings when sample size = 3000 is equal to: 9.99904
The std of mean of 1000 Poisson samplings when sample size = 3000 is equal to: 0.05912

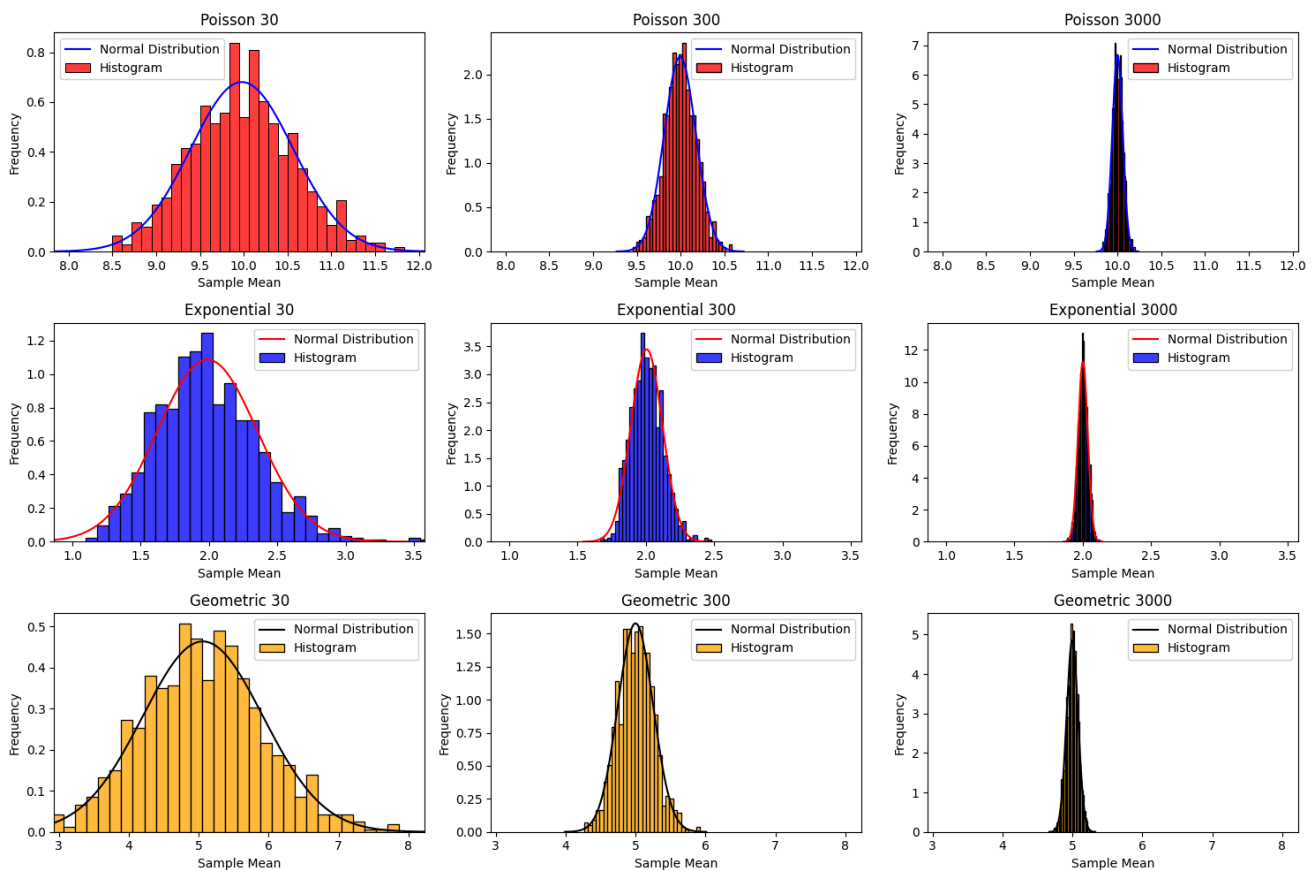
***** Exponential *****
Actual mean: 2
Actual std: 2
The mean of mean of 1000 Exponential samplings when sample size = 30 is equal to: 2.01186
The std of mean of 1000 Exponential samplings when sample size = 30 is equal to: 0.36914
The mean of mean of 1000 Exponential samplings when sample size = 300 is equal to: 2.00002
The std of mean of 1000 Exponential samplings when sample size = 300 is equal to: 0.11040
The mean of mean of 1000 Exponential samplings when sample size = 3000 is equal to: 2.00034
The std of mean of 1000 Exponential samplings when sample size = 3000 is equal to: 0.03649

***** Geometric *****
Actual mean: 5
Actual std: 4.47214
The mean of mean of 1000 Geometric samplings when sample size = 30 is equal to: 5.00553
The std of mean of 1000 Geometric samplings when sample size = 30 is equal to: 0.82064
The mean of mean of 1000 Geometric samplings when sample size = 300 is equal to: 4.99185
The std of mean of 1000 Geometric samplings when sample size = 300 is equal to: 0.26441
The mean of mean of 1000 Geometric samplings when sample size = 3000 is equal to: 5.00021
The std of mean of 1000 Geometric samplings when sample size = 3000 is equal to: 0.08057
```

As we can see, the more sample size we dedicate the better estimation we'll have for the mean of population and the less variance and std we'll have for the means(because the variance of a mean of a sampling is σ^2/n .)

Part 4: Adding Normal Distributions to The Plots And Aanalysing Them

For this part we need the pdf function for normal distribution. So I import the norm distribution from scipy library. I'll have to plot all the plots again in three loops adding the normal distribution to them.

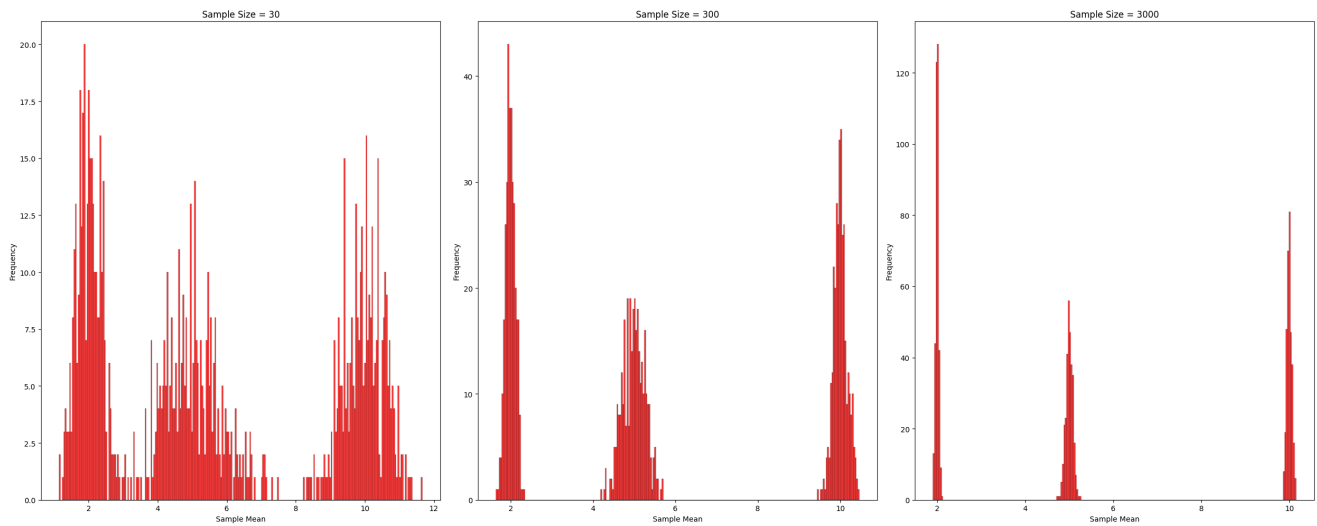


As we can see, the distribution of sample means looks more and more like the normal distribution as we increase n (the number of samples.) This is because of the CLT(Central Limit Theory) which says the more n approaches infinity the more exact estimation is the normal distribution of sum of some i.i.d RVs(which is divided by n here and is corresponding to the sample mean.)

Part 5: Choosing Distributions Randomly And Evaluating The Samplings

In this part I will randomly(with uniformly distributed probability) choose one of the three distributions and get n samples of that. I'll do this in a loop with 1000 iteration, then I'll create the sample mean list for all the samplings and try to guess their distribution by plotting them.

Now I'll plot them:



For each sample size three normal distributions get made by the data. For bigger sample sizes the normal distributions are thinner and sharper peaking around their mean.

Now I'll report their mean and std:

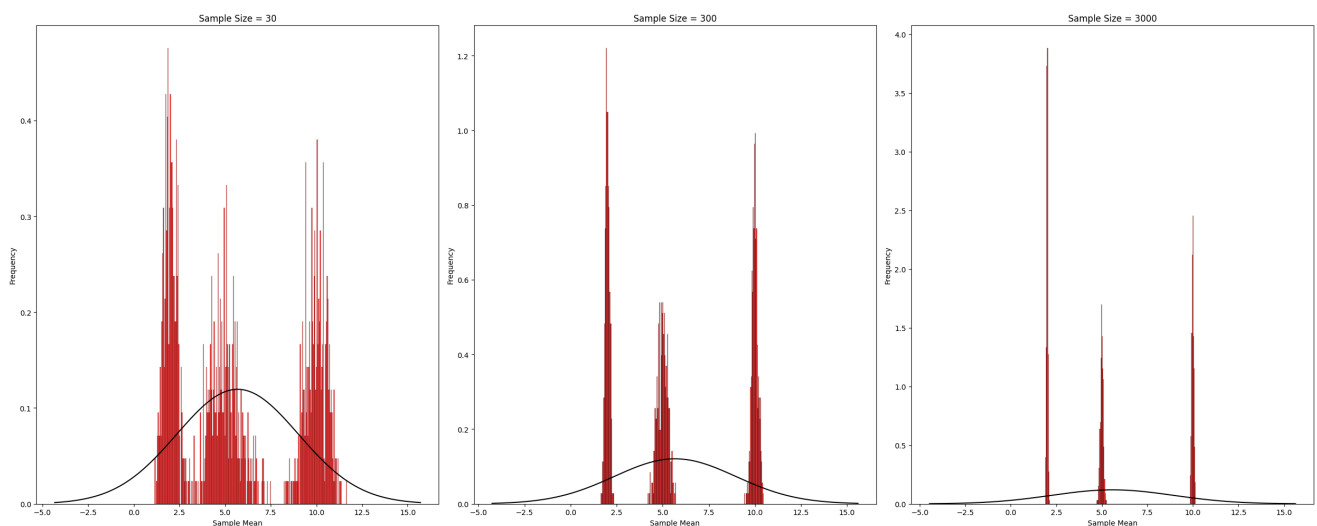
```
The mean for distribution when sample size = 30 is: 5.686295771955482
The std for distribution when sample size = 30 is: 3.338172584266237
The mean for distribution when sample size = 300 is: 5.6662216790952735
The std for distribution when sample size = 300 is: 3.309276328936072
The mean for distribution when sample size = 3000 is: 5.583761600517301
The std for distribution when sample size = 3000 is: 3.348685392995884
```

Part 6: Calculating The Mean and Std Before Sampling

The answer to this question is "No." Because the variables are chosen randomly with probability of $1/3$, so they ain't i.i.d and we can't calculate the mean and variance by only knowing the prior distributions.

Part 7: Multivariable Normal Distribution

First, I plot the normal distribution for part 5.



I suppose the distributions of part 4 are jointly normal. Because according to the CLT we know each of those 9 distributions is a normal distribution (knowing that $n \geq 30$.) And Because all of the prior variables, X_i and Y_i and Z_i , are independent from the others so the 9 distributions (which each is a normal distribution) are independent from the others. So they are jointly normal distributed; because we know that sum of some random variables from different independent normal distributions is a multivariable normal (jointly normal) distribution.